

Work Stream 3: Pre-Singularity Journey & Turbulence Modeling

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1 Introduction

As noted by u/babainottawa, while the singularity itself is physically unreachable, the *journey* of the singularity contains valuable information. The extreme vortex stretching observed in the regime where $Ma < 0.3$ offers a unique mathematical testbed for turbulence models.

2 Subgrid-Scale Stress and Singularity Trajectories

In Large Eddy Simulation (LES), the subgrid-scale (SGS) stress tensor τ_{ij} is often modeled using the Smagorinsky eddy viscosity model. By taking the exact mathematical trajectory of the OpenAI blowup construction, we can a priori test how standard SGS models respond to extreme energy cascades.

The oscillating wave packets constructed in the proof transfer energy to the core vortex in a highly structured, anisotropic manner. Analyzing this process provides a benchmark for dynamic models (like the Germano identity) under transient, non-equilibrium turbulence conditions.

3 Acknowledgments

We thank u/babainottawa for shifting the focus from the mathematical endpoint to the physical journey. *Rendons à César ce qui appartient à César.*